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MariaDB [(none)]> create database Project;
Query OK, 1 row affected (0.020 sec)

MariaDB [(none)]> show databases
-> ;

Database
company
contactsdetails
information_schema
movieco
mydb
mysql
performance_schema
prestigecars
project
sys
test

11 rows in set (0.133 sec)

MariaDB [(none)]> use project;
Database changed

MariaDB [project]> create table Employee(empNo int primary key,fName varchar(30) NOT NULL, lName varchar(30) NOT NULL, address varchar(100), DOB date, sex char(1) CHECK (sex IN('M','F')), position varchar(20) CHECK(position IN ('Manager', 'Team Leader', 'Analyst', 'Software Developer')), deptNo varchar(30));
Query OK, 0 rows affected (0.051 sec)

MariaDB [project]> describe employee;

Field	Type	Null	Key	Default	Extra
empNo	int(11)	NO	PRI	NULL	
fName	varchar(30)	NO		NULL	
lName	varchar(30)	NO		NULL	
address	varchar(100)	YES		NULL	
DOB	date	YES		NULL	
sex	char(1)	YES		NULL	
position	varchar(20)	YES		NULL	
deptNo	varchar(30)	YES		NULL	

8 rows in set (0.074 sec)

MariaDB [project]> create table Department(deptNo varchar(30) primary key, deptName varchar(50) NOT NULL, mgrEmpNo varchar(30));
Query OK, 0 rows affected (0.020 sec)

MariaDB [project]> describe Department;

Field	Type	Null	Key	Default	Extra
deptNo	varchar(30)	NO	PRI	NULL	
deptName	varchar(50)	NO		NULL	
mgrEmpNo	varchar(30)	YES		NULL	

3 rows in set (0.057 sec)

```
MariaDB [project]> alter table Employee modify empNo varchar(30);
Query OK, 0 rows affected (0.083 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [project]> describe Employee;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(30)	NO	PRI	NULL	
fName	varchar(30)	NO		NULL	
lName	varchar(30)	NO		NULL	
address	varchar(100)	YES		NULL	
DOB	date	YES		NULL	
sex	char(1)	YES		NULL	
position	varchar(20)	YES		NULL	
deptNo	varchar(30)	YES		NULL	

8 rows in set (0.027 sec)

```
MariaDB [project]> alter table Employee add constraint FK_Employee_Department
foreign key(deptNo) references Department(deptNo);
Query OK, 0 rows affected (0.038 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [project]> describe Employee;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(30)	NO	PRI	NULL	
fName	varchar(30)	NO		NULL	
lName	varchar(30)	NO		NULL	
address	varchar(100)	YES		NULL	
DOB	date	YES		NULL	
sex	char(1)	YES		NULL	
position	varchar(20)	YES		NULL	
deptNo	varchar(30)	YES	MUL	NULL	

8 rows in set (0.047 sec)

```
MariaDB [project]> alter table Department add constraint
FK_Department_Manager foreign key (mgrEmpNo) references Employee(empNo);
Query OK, 0 rows affected (0.022 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
MariaDB [project]> describe Department;
```

Field	Type	Null	Key	Default	Extra
deptNo	varchar(30)	NO	PRI	NULL	
deptName	varchar(50)	NO		NULL	
mgrEmpNo	varchar(30)	YES	MUL	NULL	

3 rows in set (0.053 sec)

```
MariaDB [project]> create table Project(projNo varchar(30) primary key,
projName varchar(50) NOT NULL, deptNo varchar(30) NOT NULL, foreign
key(deptNo) references Department(deptNo));
Query OK, 0 rows affected (0.020 sec)
```

```
MariaDB [project]> describe Project;
```

Field	Type	Null	Key	Default	Extra
-------	------	------	-----	---------	-------

projNo	varchar(30)	NO	PRI	NULL		
projName	varchar(50)	NO		NULL		
deptNo	varchar(30)	NO	MUL	NULL		

3 rows in set (0.053 sec)

```
MariaDB [project]> create table WorksOn(empNo varchar(30), projNo
varchar(30), dateWorked date, hoursWorked int check (hoursWorked BETWEEN 0
AND 40), primary key(empNo, projNo, dateWorked), foreign key(empNo)
references Employee(empNo), foreign key(projNo) references Project(projNo));
Query OK, 0 rows affected (0.022 sec)
```

```
MariaDB [project]> describe WorksOn;
```

Field	Type	Null	Key	Default	Extra
empNo	varchar(30)	NO	PRI	NULL	
projNo	varchar(30)	NO	PRI	NULL	
dateWorked	date	NO	PRI	NULL	
hoursWorked	int(11)	YES		NULL	

4 rows in set (0.036 sec)

```
MariaDB [project]> INSERT INTO Department (deptNo, deptName, mgrEmpNo) values
-> ('1', 'IT', NULL),
->
-> ('2', 'Planning', NULL),
-> ('3', 'Finance', NULL);
Query OK, 3 rows affected (0.078 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
MariaDB [project]> select * from department;
```

deptNo	deptName	mgrEmpNo
1	IT	NULL
2	Planning	NULL
3	Finance	NULL

3 rows in set (0.001 sec)

```
MariaDB [project]> insert into Employee (empNo, fName, lName, address, DOB,
sex, position, deptNo) values
-> ('101', 'Alice', 'Mensah', 'Accra', '1995-05-12', 'F', 'Manager',
'1'),
-> ('102', 'Bob', 'Owusu', 'Kumasi', '1992-03-20', 'M', 'Analyst', '1'),
-> ('103', 'Cynthia', 'Boateng', 'Takoradi', '1998-07-08', 'F',
'Manager', '2'),
-> ('104', 'David', 'Asare', 'Cape Coast', '1988-09-15', 'M', 'Team
Leader', '2'),
-> ('105', 'Evelyn', 'Addo', 'Tamale', '1991-11-25', 'F', 'Software
Developer', '3');
Query OK, 5 rows affected (0.019 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [project]> select * from Employee;
```

empNo	fName	lName	address	DOB	sex	position	deptNo
101	Alice	Mensah	Accra	1995-05-12	F	Manager	1

102	Bob	Owusu	Kumasi	1992-03-20	M	Analyst
1						
103	Cynthia	Boateng	Takoradi	1998-07-08	F	Manager
2						
104	David	Asare	Cape Coast	1988-09-15	M	Team Leader
2						
105	Evelyn	Addo	Tamale	1991-11-25	F	Software Developer
3						

5 rows in set (0.000 sec)

```
MariaDB [project]> UPDATE Department SET mgrEmpNo = '101' WHERE deptNo = '1';
Query OK, 1 row affected (0.003 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [project]> UPDATE Department SET mgrEmpNo = '103' WHERE deptNo = '2';
Query OK, 1 row affected (0.002 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [project]> UPDATE Department SET mgrEmpNo = '105' WHERE deptNo = '3';
Query OK, 1 row affected (0.014 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
MariaDB [project]> select * from Department;
+-----+-----+-----+
| deptNo | deptName | mgrEmpNo |
+-----+-----+-----+
| 1      | IT       | 101      |
| 2      | Planning | 103      |
| 3      | Finance  | 105      |
+-----+-----+-----+
3 rows in set (0.000 sec)
```

```
MariaDB [project]> INSERT INTO Project (projNo, projName, deptNo) VALUES
-> ('201', 'SCCS', '1'),
-> ('202', 'Payroll System', '1'),
-> ('203', 'Budget Planner', '2'),
-> ('204', 'Audit Tracker', '3');
Query OK, 4 rows affected (0.016 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
MariaDB [project]> select * from Project;
+-----+-----+-----+
| projNo | projName      | deptNo |
+-----+-----+-----+
| 201    | SCCS          | 1      |
| 202    | Payroll System | 1      |
| 203    | Budget Planner | 2      |
| 204    | Audit Tracker  | 3      |
+-----+-----+-----+
4 rows in set (0.001 sec)
```

```
MariaDB [project]> INSERT INTO WorksOn (empNo, projNo, dateWorked,
hoursWorked) VALUES
-> ('101', '201', '2026-01-05', 8),
-> ('101', '201', '2026-01-06', 7),
-> ('102', '201', '2026-01-05', 6),
-> ('102', '202', '2026-01-07', 8),
-> ('103', '203', '2026-01-08', 8),
-> ('103', '203', '2026-01-09', 7),
-> ('104', '203', '2026-01-10', 5),
-> ('105', '204', '2026-01-11', 8),
-> ('105', '204', '2026-01-12', 6);
```

Query OK, 9 rows affected (0.007 sec)
Records: 9 Duplicates: 0 Warnings: 0

MariaDB [project]> select * from WorksOn;

empNo	projNo	dateWorked	hoursWorked
101	201	2026-01-05	8
101	201	2026-01-06	7
102	201	2026-01-05	6
102	202	2026-01-07	8
103	203	2026-01-08	8
103	203	2026-01-09	7
104	203	2026-01-10	5
105	204	2026-01-11	8
105	204	2026-01-12	6

9 rows in set (0.001 sec)

Exercise 7.25

MariaDB [project]> create view FemaleManagedProjects as select p.projNo, p.projName FROM Project p join Department d ON p.deptNo = d.deptNo JOIN Employee e ON d.mgrEmpNo = e.empNo WHERE e.sex = 'F' ORDER BY p.projNo;
Query OK, 0 rows affected (0.020 sec)

MariaDB [project]> select * from FemaleManagedProjects;

projNo	projName
201	SCCS
202	Payroll System
203	Budget Planner
204	Audit Tracker

4 rows in set (0.008 sec)

Exercise 7.26

MariaDB [project]> create view EmployeeProjectHours as select e.empNo, e.fName, e.lName, p.projName, w.hoursWorked from Employee e join WorksOn w on e.empNo = w.empNo join Project p on w.projNo = p.projNo;
Query OK, 0 rows affected (0.017 sec)

MariaDB [project]> select * from EmployeeProjectHours;

empNo	fName	lName	projName	hoursWorked
101	Alice	Mensah	SCCS	8
101	Alice	Mensah	SCCS	7
102	Bob	Owusu	SCCS	6
102	Bob	Owusu	Payroll System	8
103	Cynthia	Boateng	Budget Planner	8
103	Cynthia	Boateng	Budget Planner	7
104	David	Asare	Budget Planner	5
105	Evelyn	Addo	Audit Tracker	8
105	Evelyn	Addo	Audit Tracker	6

9 rows in set (0.015 sec)

Exercise 7.27

MariaDB [project]> create view EmpProject(empNo, projNo, totalHours) as select w.empNo, w.projNo, SUM(hoursWorked) from Employee e, Project p, WorksOn w where e.empNo = w.empNo and p.projNo = w.projNo group by w.empNo, w.projNo;
Query OK, 0 rows affected (0.010 sec)

a)

```
MariaDB [project]> select * from EmpProject;
```

empNo	projNo	totalHours
101	201	15
102	201	6
102	202	8
103	203	15
104	203	5
105	204	14

6 rows in set (0.003 sec)

b)

```
MariaDB [project]> select projNo from EmpProject where projNo = 'SCCS';
```

projNo
SCCS
SCCS

2 rows in set (0.003 sec)

c)

```
MariaDB [project]> select count(projNo) from EmpProject where empNo = '101';
```

count(projNo)
1

1 row in set (0.002 sec)

d)

```
MariaDB [project]> select empNo, totalHours from EmpProject group by empNo;
```

empNo	totalHours
101	15
102	6
103	15
104	5
105	14

5 rows in set (0.001 sec)

Exercise 7.28

There is no need to access the Part table when a new row is inserted with partCost > 1000, a row is updated from partCost